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Introducing a method to derive an enterprise-specific SOA operating model

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Abstract

Considering that each IT-organization is based on an established IT-governance the introduction of SOA as a new overall architectural approach will change the requirements for the specific governance model. The paper presents a SOA-specific governance model which differentiates an operating model, service life-cycle management, service ownership, cost allocation, and meta-data management support. Based on the company´s maturity along the SOA-relevant dimensions IT-governance, SOA experience, and process maturity, its size, as well as its organizational structure the paper discusses an SOA operating model with organizational structure, roles, responsibilities, and KPI´s. The main assumption of the approach is the alignment of the organizational structure of IT with a functional domain model. This operating model is the outcome of ten industrial workshops and one complex case study based on standards as CobiT and ITIL.

1. Introduction

When introducing a service-oriented architecture (SOA) in an enterprise, not only the technical aspects of the architecture and of the service development are important. SOA is a new paradigm for enterprise architectures with severe consequences on the IT and business side.

"SOA is difficult to implement, manage, and control. Not because of the technology […] but due to the organizational, cultural, and behavioural aspects of SOA that contribute to success." [1]

"When talking about enterprise IT, it is important to realize that many – if not most – of the problems associated with it are not of a technical nature but can be found on the organizational level instead." [2]

An empirical study by Aberdeen Group among companies that are introducing SOA confirms this [3]. One of the main reasons lies in the highly distributed architecture. SOA is characterized by many heterogeneous, reused, and flexible services of different granularity and in different lifecycles with complicated provider-consumer relationships across organizational boundaries. Risks that arise from this situation are among others the loss of the overview over the IT system or the corruption of business processes and the IT-system due to the failure of only a few underlying services.

For an enterprise in order to reap the benefits of SOA, it is a prerequisite to tackle these challenges by performing numerous changes regarding the organization and management processes on the IT but also on the business side. Challenges are among others: providing services at defined reliable service levels, reusing services, building exactly the reusable services the consumers need, and managing risk.

The proposed SOA governance framework bundles the necessary changes and activities on the organizational level in an actionable and holistic concept. It supports companies to plan and introduce SOA successfully and manage it sustainably. The framework aims at fulfilling the SOA-specific requirements building on the existing IT governance, trying to economize on management attention and company resources.

The article is structured as follows. Firstly, the theoretical basis of SOA, of IT governance, and of SOA governance is discussed. Secondly, an overview of the SOA governance framework is given, detailing the introduction method, operating model, organizational structure, as well as management processes and guidelines. In the last part of the article, the methodology of how this framework has been developed and how it has been evaluated is described.

2. Theoretical foundation

This part describes the theoretical foundation of SOA, IT governance, SOA governance, and existing frameworks.

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2.1. SOA

There is no unique definition for the term “SOA”. There are definitions with a more technical focus and ones that include enterprise architecture aspects. We developed an own definition for SOA:

SOA is an architecture style that brings together elements of software architecture and enterprise architecture. It is based on the interaction of services. Services are autonomous as well as interoperable and provide re-usable functions via a technically standardized interface. Services can exist on all application system layers (business process, presentation, business logic, data management). They can be aggregated based on lower-level services, encapsulated from existing IT systems but also implemented from scratch.

Enterprise Architecture (EA) can be defined as “…a framework or ’blueprint’ for how the organization achieves the current and future business objectives. It examines the key business, information, application, and technology strategies and their impact on business functions. Each of these strategies is a separate architectural discipline and Enterprise Architecture is the glue that integrates each of these disciplines into a cohesive framework.“ [4] Steen et al. point out that “…the enterprise architecture combines and relates all architectures describing some aspect of the organization, such as the business process architecture, the information architecture, and the application architecture. It is a blueprint of the organisation, which serves as a starting point for analysis, design and decision making.” [5]

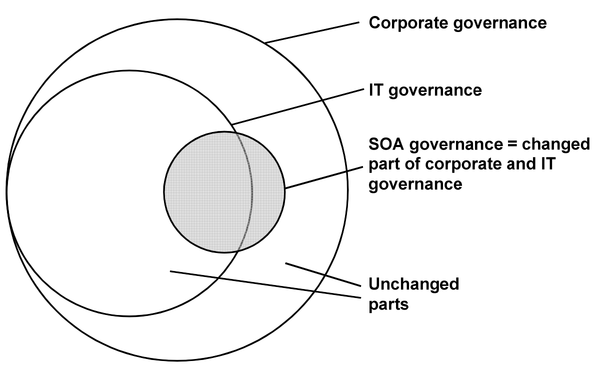

Figure 1. Relation between corporate, IT, and SOA governance

Universally recognized characteristics of SOA are furthermore the distributed structure as well as aspects of orchestration and loose coupling of services. Partial aspects of this definition can be found in the one by McCoy and Natis as well as by Marks and Bell [1, 6].

Numerous advantages are expected from an introduction of an SOA into an enterprise: re-use, cost efficiency, increased quality of the software components, flexibility, i.e., faster and simpler adaptation of the IT systems to changed business processes, increased transparency and outsourcing opportunities, lower time to market of new products or services and higher compatibility. Further goals that are relevant for an agile enterprise are a more efficient development process, the opportunity of a step-wise introduction and extension, risk mitigation, and the independence of specific technology. [1, 7] An adequate SOA governance is a prerequisite for the realization of all these goals.

2.2. IT governance

„…corporate governance could be defined as ‘the process of controlling management and of balancing the interests of all internal stakeholders and other parties (external stakeholders, governments and local communities […]) who can be affected by the corporation’s conduct in order to ensure responsible behavior by corporations and to achieve the maximum level of efficiency and profitability for a corporation’.” [8]

Due to the importance of IT for today’s enterprises, IT governance and corporate governance must be integrated [9]. IT governance is the part of governance that relates to the information and communication systems in an enterprise. The ITGI (IT Governance Institute), author of the CobiT framework (Control Objectives for Information and related Technology), defines IT governance as follows:

„IT governance is the responsibility of executives and the board of directors, and consists of the leadership, organizational structures and processes that ensure that the enterprise’s IT sustains and extends the organization’s strategies and objectives.” [10]

Peterson defines IT governance as follows:

„The distribution of IT decision-making rights and responsibilities among enterprise stakeholders, and the procedures and mechanisms for making and monitoring strategic decisions regarding IT.“ [11]

Rather than focusing on individual decisions, IT governance is concerned with influencing peoples’ behavior when using and designing IT [12].

Based on the understanding of IT governance SOA governance can be defined as follows:

„SOA governance is more than providing governance for SOA efforts; it is how IT governance should operate within an organization that has adopted SOA as their primary approach to enterprise architecture.“ [13]

2.3. IT and SOA governance frameworks

The introduced SOA governance framework is based on two foundations: IT governance literature and SOA-specific management literature.

According to van Grembergen's holistic framework three components belong to IT governance: structures, processes, and relational mechanisms.

„Structures involve the existence of responsible functions such as IT executives and accounts, and a diversity of IT committees. Processes refer to strategic IT decision-making and monitoring. The relational mechanisms include business/IT participation and partnerships, strategic dialogue and shared learning." [9]

Ross und Weill stress the distribution of decision making rights in their definition and present a „Governance Design Framework" by MIT Sloan School for IS Research with the following components [12]

- Enterprise strategy and organization
- IT organization and desirable behavior
- IT governance arrangements
- IT governance mechanisms (with decisions related to principals, architecture, infrastructure, and investment)
- Business performance goals
- IT metrics and accountabilities

The aspects mentioned by van Grembergen and Ross/Weill are reflected in the four components of the presented SOA governance framework: operating model and organizational structure (with roles and responsibilities), management processes and guidelines, collaboration and communication, as well as performance management (cf. Figure 3).

CobiT is a framework for implementing IT governance in an enterprise [10]. 34 processes within the four process areas „Plan and Organize", „Acquire and Support", „Deliver and Support", and „Monitor and Evaluate" are documented. Every process documentation contains the activities with their inputs, outputs, RACI matrices (Responsible, Accountable, Consult, Inform), as well as goal and performance indicators. These processes shall ensure an effective management of the information and of the IT resources. [14]

ITIL (IT Infrastructure Library, currently version three) by the ITSMF (IT Service Management Forum) describes best practices for IT management processes and is an established standard [15].

The presented SOA governance framework focuses on those ITIL and CobiT processes that are subject to SOA-specific adaptations. These targeted changes are easier to implement for companies that have a working IT governance (CobiT) and adequate IT service management.

As mentioned above, the problem of managing an SOA, i.e., SOA governance, is discussed in literature. In [16] a "Human Service Bus" is described as the appropriate organizational setup in a company that introduces SOA. The article proposes a domain-based organizational structure. In domains, services that fulfill similar business requirements are clustered. This grouping can be used for taking decisions and performing management activities. The article elaborates on how employees can be motivated by individual incentives. It constitutes a good foundation for our presented framework. However, it does not offer concrete design proposals. In [17], some of the SOA-specific roles are documented without metrics.

[18] is a whitepaper that focuses on the importance of SOA guidelines, service contracts, meta-data management, and service lifecycle management. However, beyond motivation to deal with these issues, no further detailed design proposals or guidelines are given.

[19] focuses on the following partial areas of SOA governance: service definition, service deployment lifecycle, service versioning, service migration, service registries, service data models, service monitoring, service properties, service tests, and service security. Partially, solution approaches are discussed. These are included in our work. [13] describes on a high level architecture, business, data, and technical principles, that should be used in an SOA. It shows which architecture management processes are relevant and describes the domain principle.

The Open Group (www.opengroup.org/projects/soa-togaf) works on how TOGAF 8 (The Open Group Architecture Framework) can be used for managing an SOA. First results are discussed in [20].

3. SOA governance framework

Topics relevant to SOA governance are enterprise architecture, business process, service management, initiative management, provider management, and infrastructure management (cf. Figure 2). The proposed SOA management framework focuses on the two core areas architecture and service management since they contain the main SOA-specific changes. Important tasks of architecture management are developing an SOA strategy, architecture maps, and architecture roadmaps as well as defining architecture standards and monitoring compliance. Service management comprises the management of the services along their full lifecycle and the management of service levels and SLAs (service level agreements).

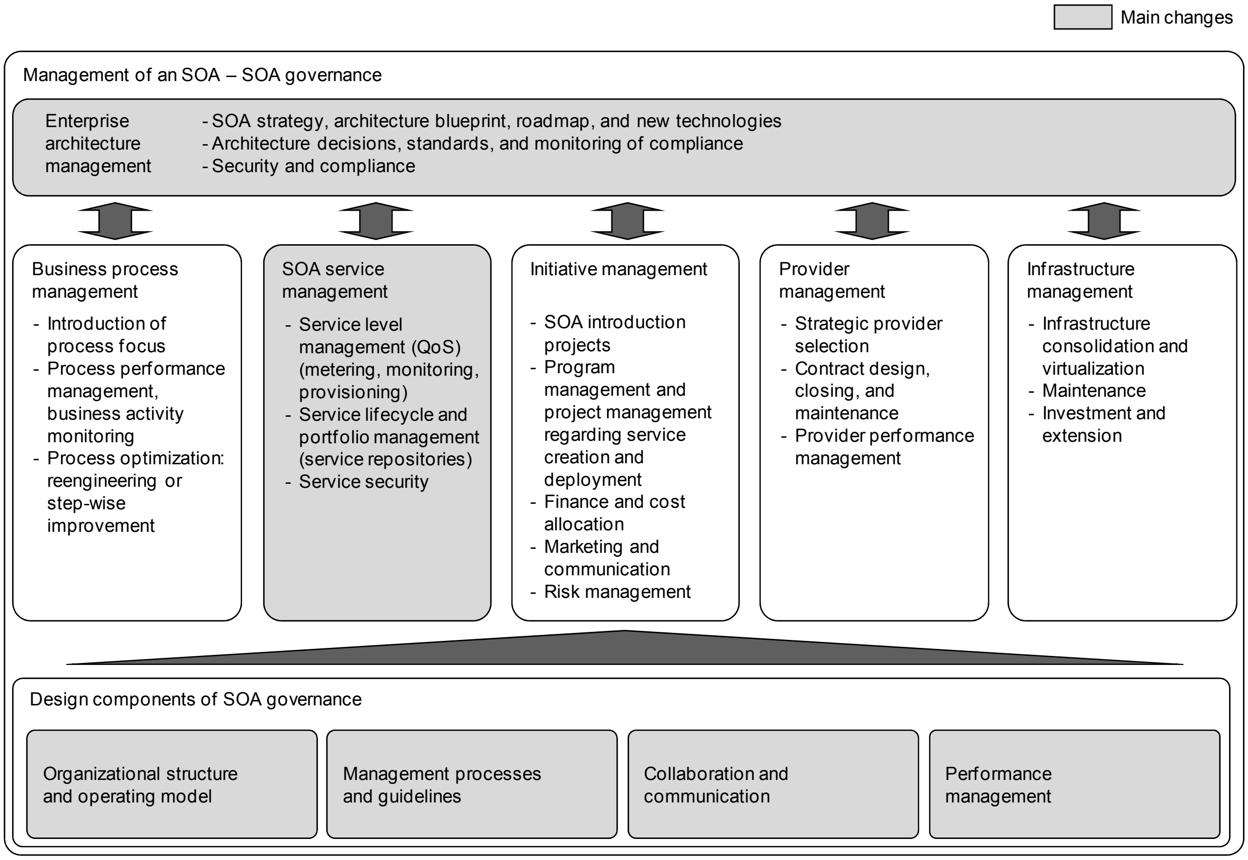

Figure 2. Overview of the aspects of SOA governance

The proposed framework has two major parts: design proposals for a SOA governance concept that covers the above mentioned requirements as well as a method for implementing SOA governance that takes into consideration the maturity level of the respective enterprise.

3.1. Method for implementing SOA governance

The method for implementing an SOA governance within an enterprise proposed in this paper consists of the following steps: “Define SOA strategy”, “Analyze maturity”, “Design SOA governance concept”, “Implement SOA governance”, as well as “Optimize and manage on ongoing basis”. First of all, the SOA introduction team is assembled and the goals, the strategy, as well as the rough roadmap are developed. In the analysis phase, the SOA introduction team assesses the maturity of the IT governance and architecture, the change readiness, as well as the current situation of the company.

In the design phase, based on the results of the first two phases, the team adapts the generic SOA governance concept provided by us at the specific situation in the enterprise and documents the outcome.

In the planning phase, the SOA introduction team breaks down the implementation of the developed concepts into tasks and puts them on a timeline with concrete milestones. Training material must be developed. In the next phase, the implementation phase, the employees are trained and the implementation tracked with the implementation metrics. The last phase is about optimizing the concept, adapting it to changed situations, and monitoring the performance of the processes.

3.2. Overview of proposed SOA governance design artifacts for the design phase

All described design artifacts are generic proposals that contain specific changes to the current IT governance when introducing an SOA. The design artifacts were developed based on literature study and expert interviews as well as evaluated and improved in workshops and a case study. They are not intended to describe mandatory target structures and mechanisms but rather support the discussions geared towards finding the right enterprise-specific SOA governance model.

In this article, we focus on the operating model and organizational structure with roles and responsibilities as well as the management processes and guidelines. Due to space limitations, collaboration and communication, performance management, as well as tool support are only mentioned briefly.

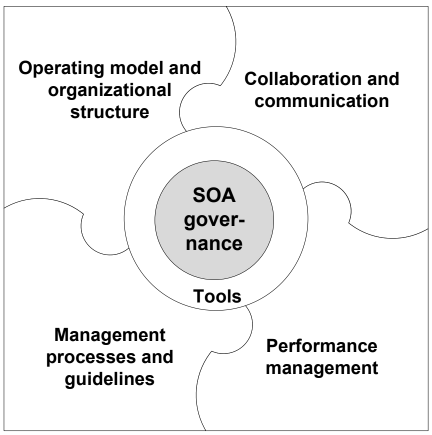

Figure 3. Components of designing SOA governance (based on [9])

3.3. Organizational structure and operating model

In this context, the following types of changes are potentially necessary: the adaptation of the operating model and the organizational structure on the business and IT side, the introduction of virtual SOA structures, as well as the introduction of SOA-specific responsibilities for existing roles and of new SOA-specific roles.

The necessary changes to the operating model and therefore to the organizational structure are very much depending on the current situation of the enterprise. During the work on the SOA governance framework and its evaluation, five typical situations have been identified:

- Large IT service provider without business demand IT organization
- Large IT service provider with a business demand IT organization
- Large IT service provider without a business demand IT organization, which serves internationally distributed locations with similar business
- Large IT service provider without a business demand IT organization which serves nationally distributed locations with the same business
- Very small IT unit without prior SOA experience

We understand the term "business demand IT organization" as a thin IT department that elicits and bundles the IT requirements of the business side, translates them into technical requirements, and that manages the internal and external IT departments/providers that implement, provide, and run the services.

Taking the large IT service provider without a business demand IT organization as example, we describe a SOA operating model (cf. Figure 4).

The following design goals were considered:

- Functional expertise on the business and IT side as well as reuse of services and other artifacts
- Process focus
- Service orientation
- Business as driver for IT functionality
- Technical know-how on the IT side
- Support of the business strategy and flexibility regarding the business model of individual departments by IT

The final target structure of the operating model should contain a kind of a matrix organization with functional units but also a process focus on the business side, grounded in the design goals functional expertise and process focus.

The functional expertise necessitates a functional structure on the business side with the goals of specialization and building up competencies. This type of structure is therefore widely realized in today's companies. The functional specialization can for example be made based on product lines, service lines, or customer groups. Heads of Lines of Business (I) lead the resulting functional divisions.

On the IT side too, technical know-how and specific functional knowledge necessary for building the IT functionality supporting the business should be built up. That increases development efficiency. The SOA services should be reused across the organization (see broad arrows in Figure 4). Therefore the guiding principle for the structure of the IT organization should be functional domains, i.e., functional IT domains and sub domains where necessary due to size. Functional domains are derived from the business architecture, i.e., business processes and business capabilities. However, they do not necessarily correspond to the organizational structure on the business side. There are horizontal and vertical IT domains as well as an infrastructure service domain. Vertical domains bundle services with specific business functionality that are useful for a part of the organizational units on the business side. Horizontal domains bundle shared services on the business level. The infrastructure domain offers infrastructure services, e.g., regarding IT resource provisioning, security, and monitoring.

IT has to support the business strategy of the divisions and departments. At the same time the technical possibilities in the IT influence business opportunities. The domains are a good framework for discussing about these mutual consequences, hence IT strategy should be derived and defined per IT domain based on the business strategy.

IT domain leaders (II in Figure 4) lead the IT domains. They are responsible for the provisioning of all IT services of their domain, including the SOA services. They report to the CIO. The service managers (III) that report to the domain leader are responsible for a smaller group of services within a domain. Assuring the functionality and non-functional quality of service as well as their management along the whole lifecycle are their main tasks.

The process focus plays an important role in an SOA. Therefore, process owners are introduced on the business and IT side of the organization. The process owner on the business side (IV) is responsible for guaranteeing the operation and for optimizing a group of business processes (business process management). This is the basis for using the flexibility offered by SOA and thus creating a real business benefit. The IT process owner (V) supports the business process owner in realizing changes technically or finding errors using his technical know-how about the service interfaces and the orchestration.

Process owners are responsible for processes that run within and across domains. We understand that in some companies it is a special challenge to introduce the process focus. Problems can be a weak standing of the process responsible compared to the functionally oriented positions and the loss of position within a hierarchy. The business process owners belonging to a central business process management unit shall report to the CEO. The IT process owners should report to the CIO as part of the IT process unit.

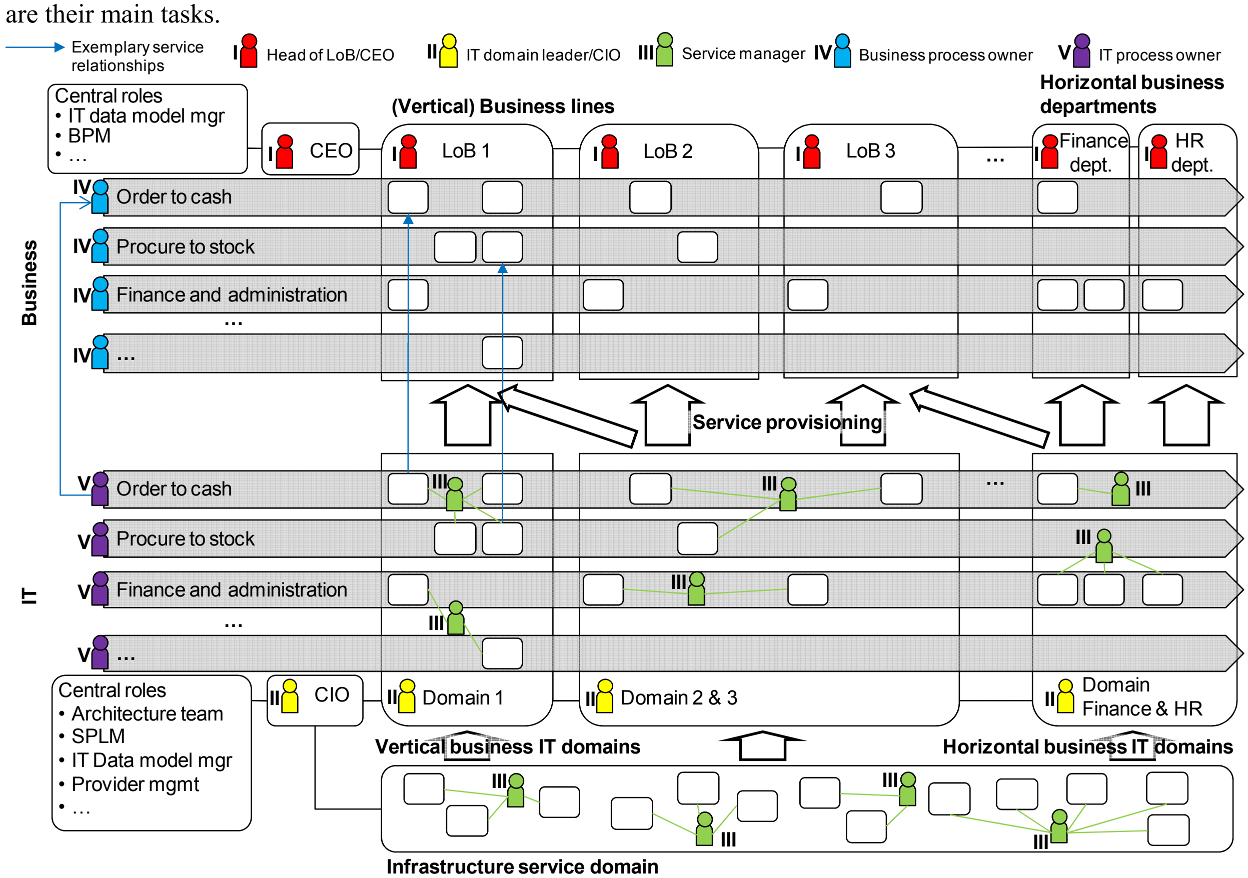

Figure 4. Overview of proposed operating model in an SOA

According to the design goals "service orientation" and "business as driver for IT functionality" business process owners and their functional counterparts are the source for requests of new functionality. The IT department is the service provider fulfilling these requests. Business and IT sign a contract for developing and operating IT process support and the services needed. This relation should be reflected in the cost allocation scheme.

A key challenge for introducing new paradigms and new technologies into an organization is the development and migration of competencies. The additional resources in an enterprise are often limited. [21] Therefore, building up new units is difficult. The concept of the SOA expert community as a virtual unit accommodates this. Employees from the central architecture team but also from the domains with interest in SOA topics and with SOA expertise are assigned to this community. These employees develop and exchange service, architecture, and technology-related knowledge in workshops, formal and informal meetings. In order to facilitate this, they must get allocated some time that they do not spend on projects. The community should lower barriers for communication and knowledge dissemination. This is done by strengthening interpersonal relationships but also by incorporating the knowledge dissemination as an important part of the promotion mechanism. The SOA experts influence the architecture of the solution across the whole lifecycle of the project. They pass expertise from one project to the next and contact other experts if they are lacking specific expertise.

Numerous roles must be introduced and the responsibility of existing ones changed – depending on what roles are already existing in the company. The roles service manager, IT domain leader, business/IT process owner, service portfolio and lifecycle management (SPLM), central architecture team, and architecture review board are especially important. Table 1 gives an overview of the most important new and changed roles. For the new roles, role descriptions have been developed, detailing goal, reported manager, KPIs, used tools, used capabilities, collaboration in groups, number of employees, and responsibilities (as RACI matrices).

Further governance bodies and roles, that exist already and that have a slightly changed field of responsibility are business division directors, IT strategy committee, IT steering committee, provider management, project manager, business analyst, software developer, security specialist, deployment manager, testing specialist, business application managers, and system/database administrators.

Table 1. New and changed roles

Role	Main tasks	KPIs
IT domain leader	Managing the vertical, horizontal, respectively infrastructure domain, its SOA services, applications, employees, finance, and projects	Service reuse, SLA compliance of services, user satisfaction with the services
Service manager	Managing a group of services within a domain; contact person for the business regarding IT functionality; negotiating SLAs and ensuring compliance	Service reuse, SLA compliance of services, user satisfaction with the services
Business process management	Supporting business processes modeling activities with methodological know-how; organizational home of the business process owners	-
Business process owner	Improving the performance of business processes	Process-specific KPIs, as cycle time
IT process owner	Mediating between business process owners and service managers; solving technical problems with the process	Process-specific KPIs on technical level, e.g., error rate
Architecture review board	Controlling the projects regarding compliance of the architecture and technology guidelines; approving the service design	-
Central architecture team	Developing IT architecture and technology guidelines; taking part in domain projects	-
Service portfolio and lifecycle management	Coordinating service requests and service versioning; managing the SOA service portfolio; controlling the quality of the service meta-data; supporting reuse	-
Data model manager	Developing and managing the extension of an enterprise-/ domain-wide logical data model	Consistency and coverage of logical data model
Technical process manager	Extending the modeled business processes by error correction, security aspects, and transactionality	Process availability, process error rates
Software architect	Defining the architecture of the IT systems to be developed according to the guidelines; defining data types, interfaces, and message patterns of the services	Quality of the architecture, service reuse, service quality

3.4. Management processes and guidelines

The SOA governance framework defines many management processes for the above mentioned roles. It also contains guidelines for all SOA-specific areas of the architecture and service management, e.g., for the communication, access to services, control mechanisms, and the behavior of the employees:

Architecture management

- Develop functional domain model
- Map IT application landscape
- Plan architecture roadmap
- Prioritize service and projects strategically
- Manage logical data model

SOA service management

- SOA service versioning and change management
- Plan and negotiate service levels
- Monitor service levels
- Implement, test, and deploy services
- Source SOA services from external providers

Additionally, certain management activities need to be included into the service design method, e.g., for testing or for searching reusable services, code snippets, requirements, and architectures, service design reviews, actively considering requirements of other potential service users, requirements prioritization and aggregation, testing, as well as negotiating and concluding a formal contract about service development and operations.

The SOA service versioning and change management process is depicted in Figure 5 as an example in a proprietary Eclipse Process Framework notation (http://www.eclipse.org/epf/). It is an important part of SOA service lifecycle management. The latter has the tasks coordinating and planning all service-related activities along the lifecycle: identification and planning, development of concept, requirements definition, service design, implementation, introduction and deployment, operations, phase-out, as well as updating and migrating services.

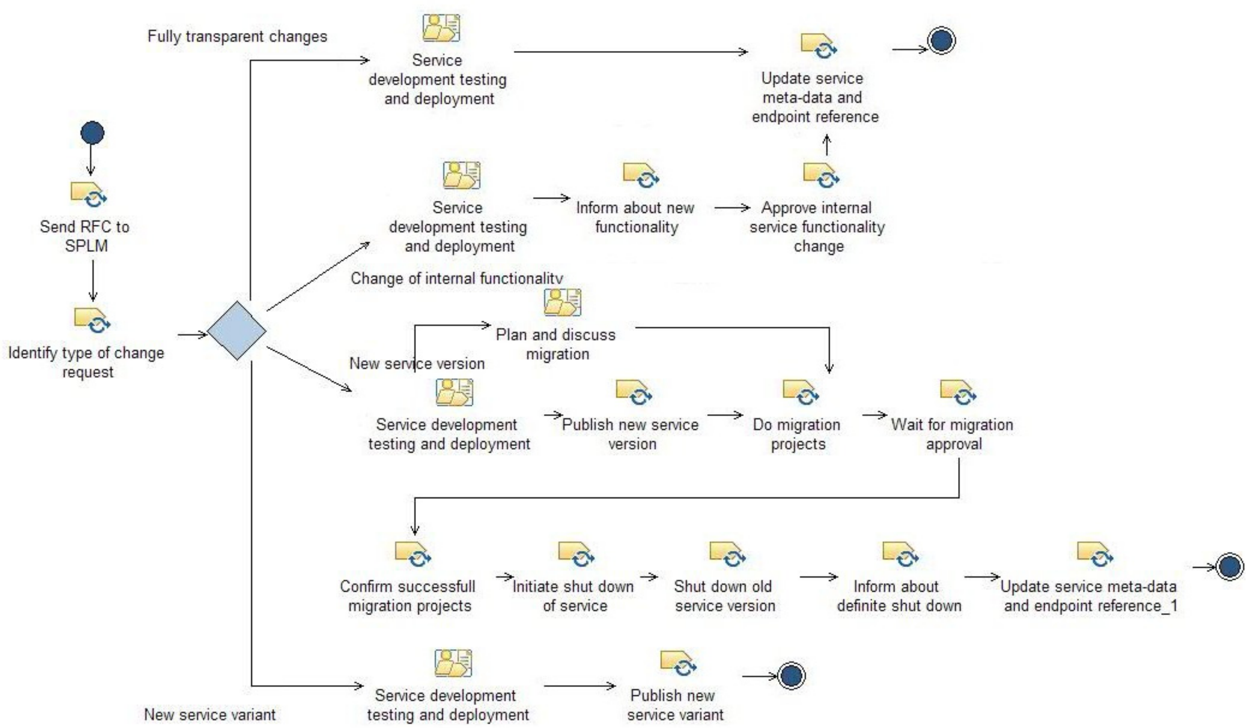

Figure 5. Process “SOA service versioning and change management”

The SOA service versioning and change management process describes how existing services can be changed while maintaining the stability of the overall IT system, e.g., not jeopardizing existing service usage relations, by using service versioning and variants. The process is started after service design when services with similar functionality have been identified, i.e., whenever changes to existing services are potentially relevant. These changes include technical changes and optimizations. KPIs for measuring the performance of this process are among others percentage of changes, that have gone through the process, speed with which the decisions about the type of change were taken, as well as the number of successfully and timely delivered migration projects.

Four main cases have to be distinguished:

- Fully transparent changes for users, e.g., optimization on the technical level. No functional changes that are relevant to current users. In this case, the changed service merely has to be developed, tested, deployed, and updated in the repository.
- Changes of internal functionality: Changes of functionality, that are relevant to current consumers, but without changes of the interface, e.g., error correction or changed calculation. In this case, the users have to be informed about the new functionality and have to approve.
- New service version: changes of functionality that are relevant to existing service consumers with changes to the interface. The new functionality (new version) should replace the old version over time. Existing service consumers have to migrate to a newer version after some versions (e.g., three versions that they have skipped). The migration period must be agreed and migration projects must be finished successfully before the old service can be shut down.
- New service variant: change of the functionality of the service due to new requirements that are only relevant and acceptable to new or a part of the old service consumers, e.g., when introducing a new product. That is why the old service version must be maintained and cannot be shut down. The new one is a variant of the old one and can be developed, tested, deployed, and updated in the repository as a new service. Both coexist.

Further components of "Management processes and guidelines" are guidelines and decision templates. Guidelines describe behavior and the sequence of activities. Decision templates describe which decisions have to be taken when, by whom, how fast, and under consideration of which information and criteria.

3.5. Further components

The other components of the framework "collaboration and communication", "performance management" and "tool support" (cf. Figure 3) are only covered briefly in this article.

Poor collaboration and communication between business departments and IT are a common reason for problems. The communication and thus effectiveness and efficiency of the IT support can be improved by regular workshops between business and IT, e.g., discussing about feedback and experiences about IT systems/services.

Performance management is about monitoring the performance of the defined management processes as well as their improvement and the compliance with rules by measuring the achievement of KPI targets for the roles and processes. Metrics and targets have been derived from CobiT (cf. [10]). Further important aspects of performance management are the clarification of the service ownership question and how costs should be distributed. For both, the framework proposes design options that are compliant with common SOA targets. A just cost distribution according to cause can have a considerably positive influence on the behavior of stakeholders and their decisions [16].

The introduction of management processes, guidelines, performance management, as well as meta-data management must fail without adequate tool support due to low acceptance level and efficiency. Service repositories are important for the meta-data management regarding services and other artifacts as well as the management processes on top of them. The framework describes which information regarding the service lifecycle, the service signature, the service levels, but also the processes and other artifacts should be managed in the repository. A pragmatic service level formalization for the management of non-functional properties has been developed.

4. Evaluation in workshops and case study

The theoretical literature study and expert interviews are the basis of the proposed framework. After the initial development, we evaluated the framework in eight workshops in companies that either were already introducing SOA or were planning to do so. The evaluation in workshops was conducted as follows. In a meeting before the actual workshop the goals, strategy, and roadmap of the SOA introduction in the company as well as the individual requirements were discussed. Furthermore, we have analyzed the maturity of the architecture and the IT governance in the enterprise. In the workshop – after individual preparation – an adequate operating model, organizational structure, roles, responsibilities, management processes, a cost distribution concept, and performance metrics were discussed.

After the workshops and step-wise improvement of the framework, the evaluation was continued with a case study. Within a 6-months project for a European utility company accompanying a SOA implementation, a SOA governance concept has been developed based on the framework's method and proposed concepts. The feedback and the experience have been used for further improvement.

5. Conclusion

In this article a framework for managing the SOA in an enterprise is proposed. It comprises the five components operating model and organizational structures, management processes and guidelines, communication and collaboration, performance management, and tool support for the SOA-specific areas enterprise architecture management and SOA service management.

The framework differentiates itself from the existing frameworks. Existing IT governance frameworks do not consider the SOA-specific challenges. Existing SOA governance approaches are mainly explaining the need for SOA-specific management and are covering these adaptations only on a very high level. This framework describes a coherent and integrated approach for introducing SOA governance. In particular, the framework considers the maturity of the enterprise where SOA governance should be introduced and gives concrete design proposals for the relevant challenging areas of SOA governance.

6. References

[1] E. A. Marks and M. Bell, *Executive's Guide to Service-Oriented Architecture*. Hoboken, New Jersey: John Wiley & Sons, Inc., 2006.

[2] D. Krafzig, K. Banke, and D. Slama, *Enterprise SOA – Service-Oriented Architecture Best Practices*. Upper Saddle River, New Jersey: Prentice Hall Professional Technical Reference, 2005.

[3] P. S. Kastner and R. Saia, "Management and governance: Planning for an optimized SOA application lifecycle," Aberdeen Group, Inc., Boston 2007.

[4] C. M. Pereira and P. Sousa, "A Method to Define an Enterprise Architecture using the Zachman Framework," in *ACM Symposium on Applied Computing (SAC '04)*, Nicosia, Cyprus, 2004, pp. 1366-1371.

[5] M. W. A. Steen, D. H. Akehurst, H. W. L. t. Doest, and M. M. Lankhorst, "Supporting viewpoint-oriented enterprise architecture," in *8th IEEE International Enterprise Distributed Object Computing Conference 2004 (EDOC 2004)*, 2004, pp. 201-211.

[6] D. McCoy and Y. Natis, "Service-Oriented Architecture: Mainstream Straight Ahead," in *Gartner Research*, 2003.

[7] D. Woods, *Enterprise Service Architecture*. Cambridge: O`Reilly & Associates, 2003.

[8] J. J. d. Plessis, J. McConvill, and M. Bagaric, *Principles of contemporary corporate governance*. Cambridge: Cambridge University Press, 2005.

[9] W. van Grembergen, *Strategies for information technology governance*. London: Idea Group Publishing, 2003.

[10] IT Governance Institute, *CobiT 4.1 - Framework, Control Objectives, Management Guidelines, Maturity Models*. Rolling Meadows, IL: IT Governance Institute, 2007.

[11] R. Peterson, "Crafting Information Technology Governance," *Information Systems Management Journal,* vol. Fall, 2004.

[12] J. W. Ross and P. Weill, "IT Governance: How Top Performers Manage It Decision Rights for Superior Results," Boston, Massachusetts: Harvard Business School Press, 2004.

[13] J. Bloomberg, "SOA Governance – IT Governance in the Context of Service Orientation." vol. 2006, Zapthink, Ed. Waltham, MA, 2004.

[14] M. Simonsson and C. Schröpfer, "Setting the IT organization goals," in *Enterprise Architecture - Models and analysis for information systems decision making*, P. Johnson and M. Ekstedt, Eds.: Studentlitteratur, 2007, pp. 153-212.

[15] Office of Government Commerce (OGC), *ITIL Lifecycle Publication Suite Books*: TSO (The Stationary Office), 2007.

[16] N. Bieberstein, S. Bose, L. Walker, and A. Lynch, "Impact of service-oriented architecture on enterprise systems, organizational structures, and individuals," in *IBM Systems Journal*. vol. 44, 2005, pp. 691-708.

[17] N. Bieberstein, S. Bose, M. Fiammante, K. Jones, and R. Shah, *Service Oriented Architecture Compass: Business Value, Planning, and Enterprise Roadmap*: IBM Press, 2006.

[18] Systinet, "SOA Governance: Balancing Flexibility and Control Within an SOA." vol. 2006 Burlington: Systinet, 2006.

[19] B. Woolf, "Introduction to SOA Governance - Governance: The official IBM definition, and why you need it." vol. 2006: IBM, 2006.

[20] A. Dico and D. Hornford, "Delivering SOA with TOGAF," 2008.

[21] J. M. Hiatt and T. J. Creasey, *Change management: the people side of change*. Loveland, Colorado: Prosci Research, 2003.